

**B**

BUYMORE

An Ecommerce Recommendation System

Major Project Submitted in Partial
Fulfilment for MCA Degree

Submitted By

Ahana Goswami – 13071024002

Ronnie Sarkar – 13071024006

Sayantani Basak – 13071024012

Rashi Aswani – 13071024030

Srijit Ghosh – 13071024041

Group ID - 1307102604

**B**

Problem Statement

- The challenge is to develop an intelligent e-commerce platform that provides personalized product recommendations, efficient customer support, and improved user engagement using Artificial Intelligence.



BUYMORE is an AI-powered ecommerce platform that provides:

- Personalized product recommendations.
- AI chatbot assistance.
- Secure OTP authentication.
- Wishlist and cart management.
- Admin monitoring and analytics.

Proposed Solution

Project Objectives



Develop a secure
ecommerce web
application

Implement an AI-
based
recommendation
system

B

Create separate
User and Admin
modules

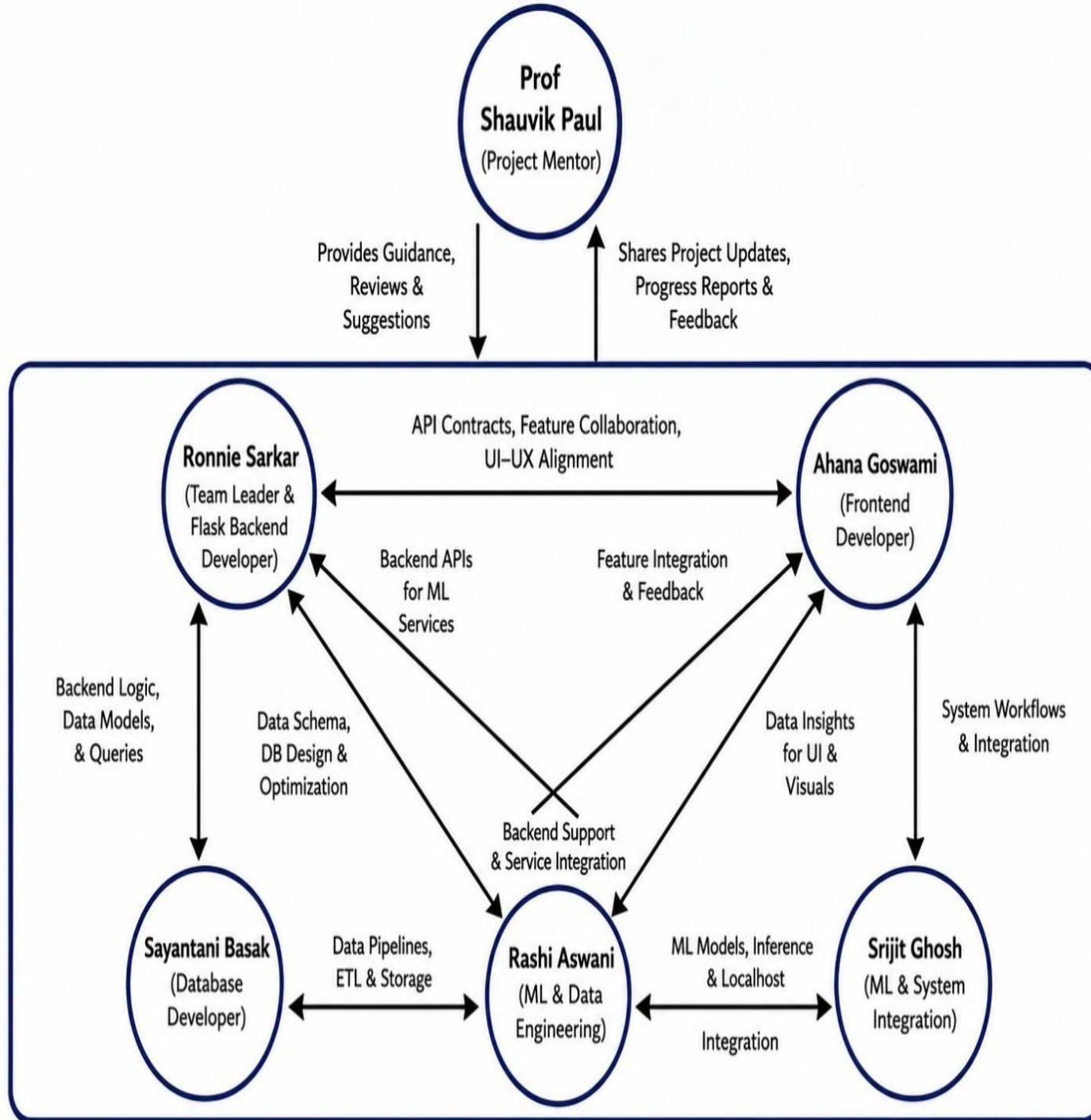
Integrate chatbot
assistance for
customers

B

Improve user
engagement using
Artificial
Intelligence

Ensure secure
database and
session handling

Team Structure



Category

Technology

Frontend	→	HTML, CSS, JavaScript
Backend	→	Flask
Database	→	MongoDB Atlas
Machine Learning	→	TF-IDF, Cosine Similarity
NLP	→	NLTK, Keras
Security	→	OTP + bcrypt
API	→	REST API



mongoDB



Technologies Used



System Features

1

User Registration & Login

2

OTP Verification

3

AI Recommendation System

4

AI Chatbot

5

Cart & Wishlist

6

Product Reviews & Ratings

7

Admin Dashboard

8

Product Analytics

9

Responsive User Interface

10

Secure Session Handling





User Module

1. User Registration

2. Login using
OTP verification

3. Browse Products

4. Search & Filter
Products

5. Add Products
To Cart

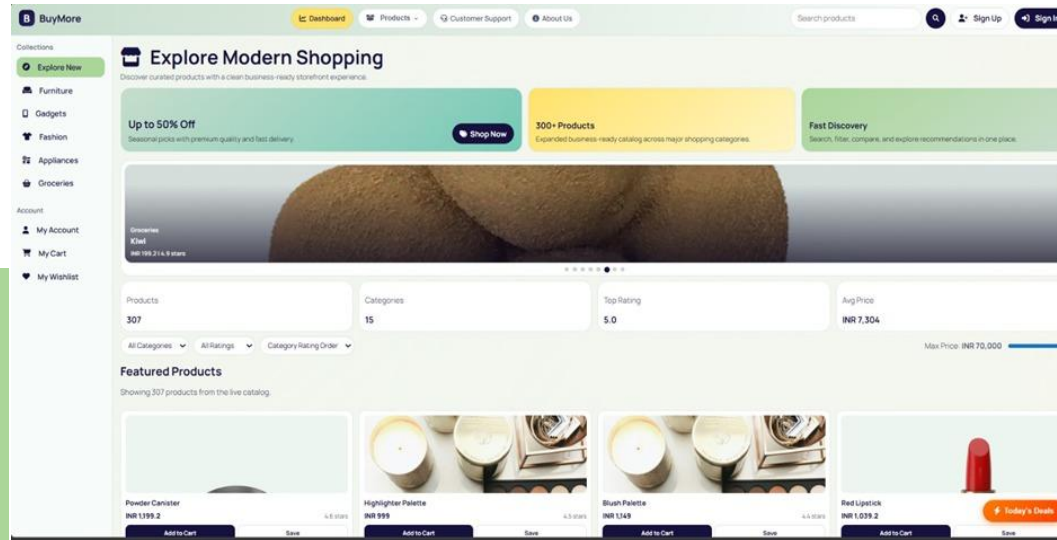
6. Add Products
To Wishlist

7. Place Orders

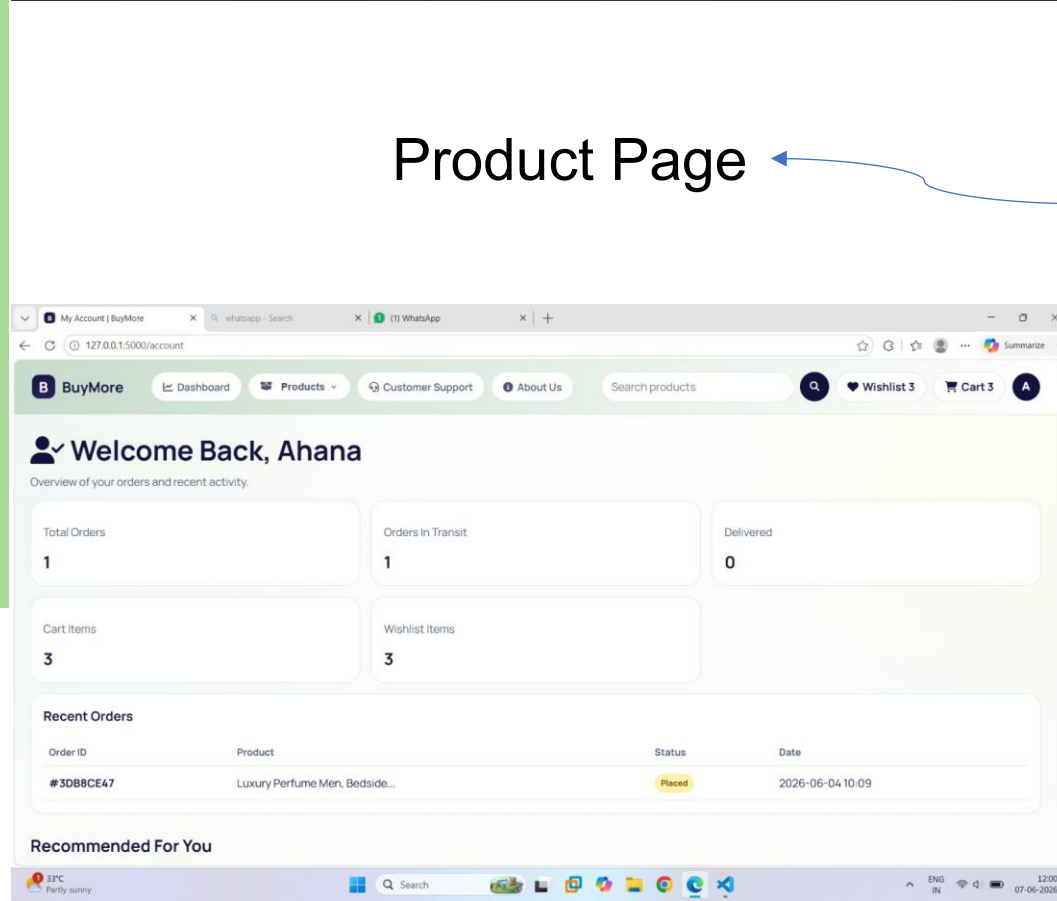
8. Submit Reviews
& Ratings

9. Receive
Personalised
Recommendation

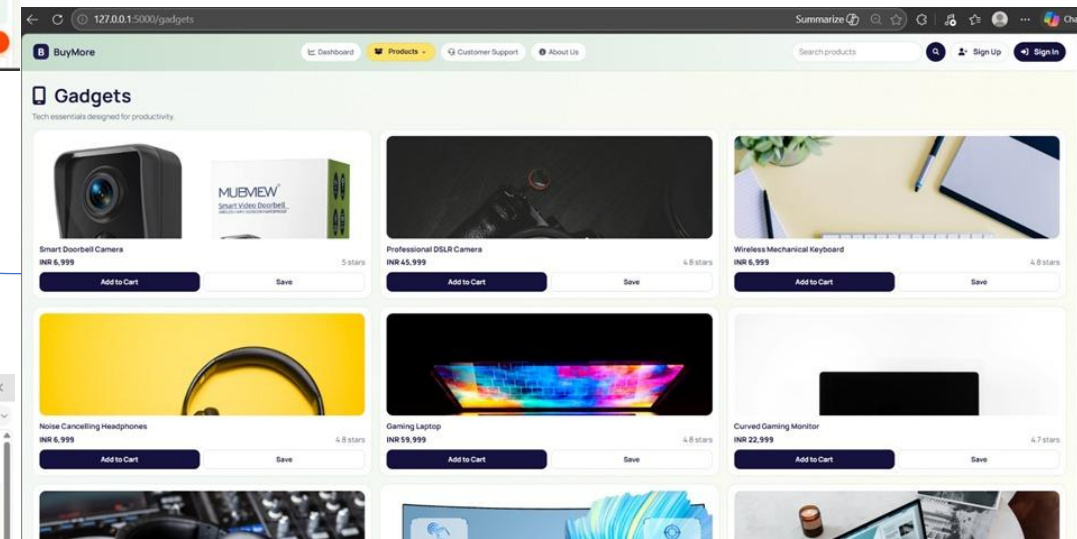
User Module Display



Home Page

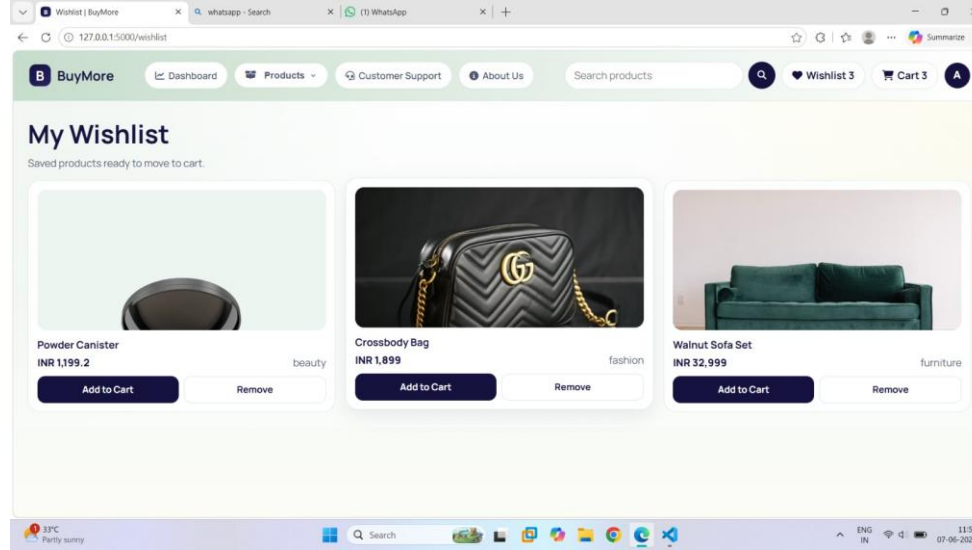


Product Page



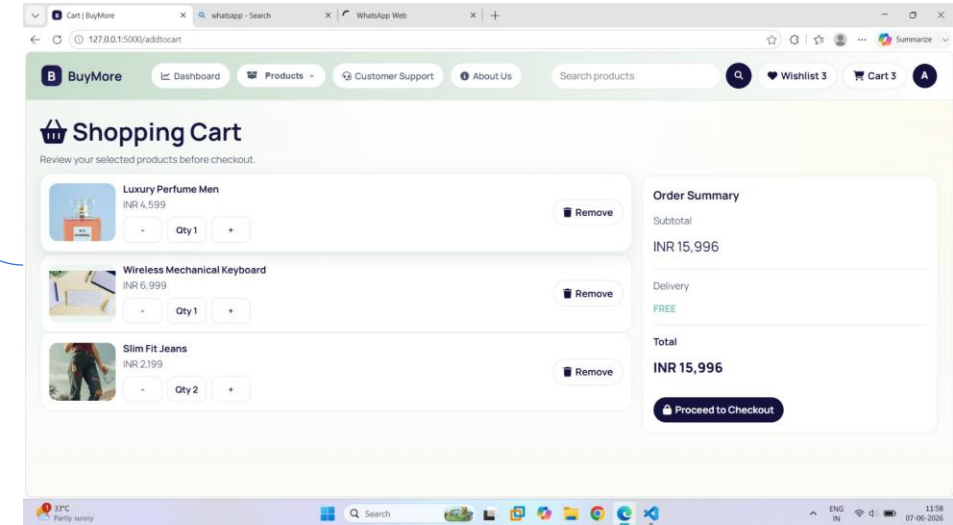
User Profile Page

User Module Features

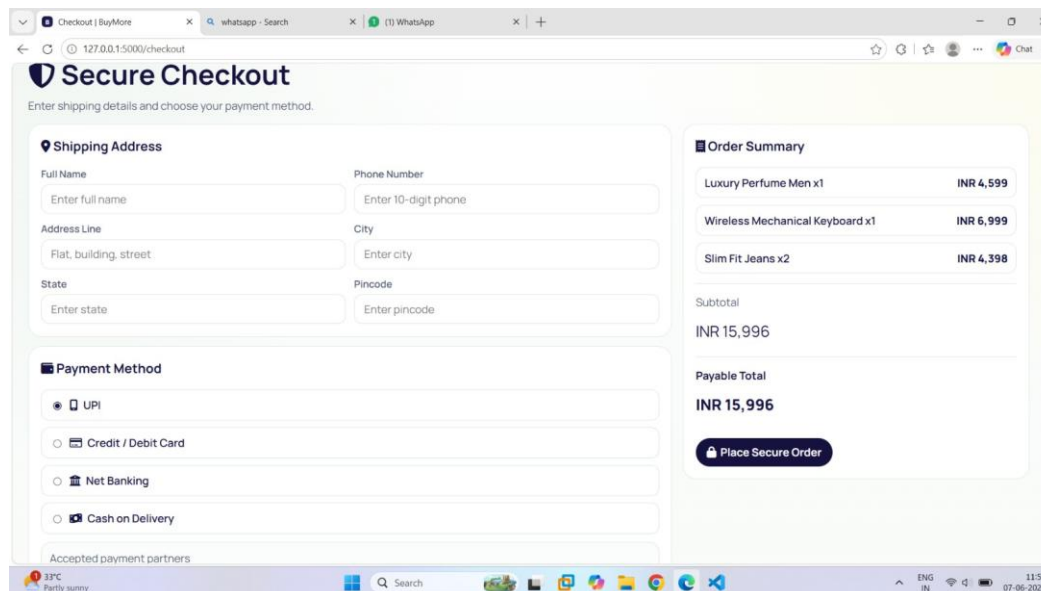


Wishlist Page

Cart Page



Checkout Page





Admin Module

Admin Login

1.

Add / Update /
Delete Products

2.

Monitor User Activity

3.

Manage Orders

4.

Inventory
Management

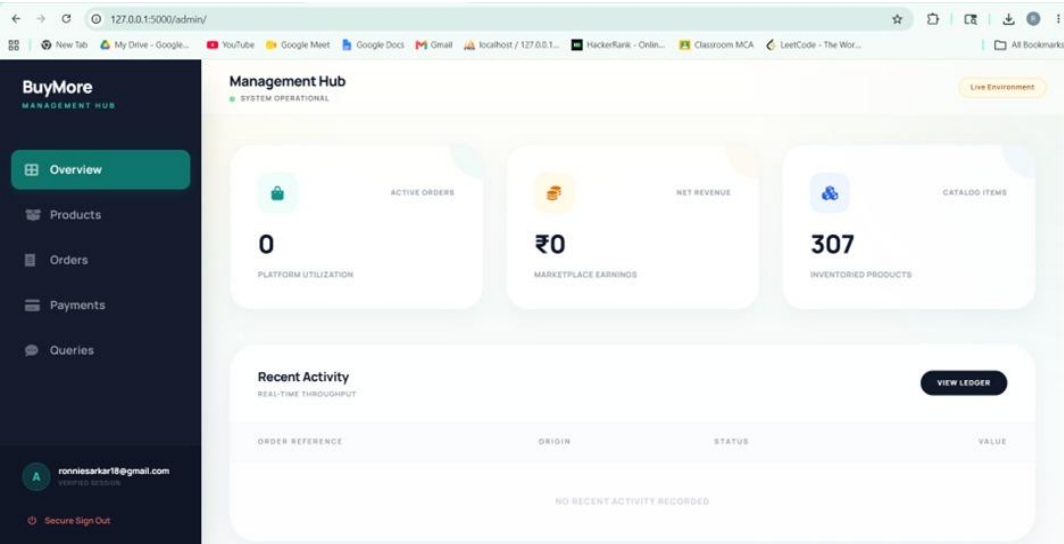
5.

View Product
Analytics

6.

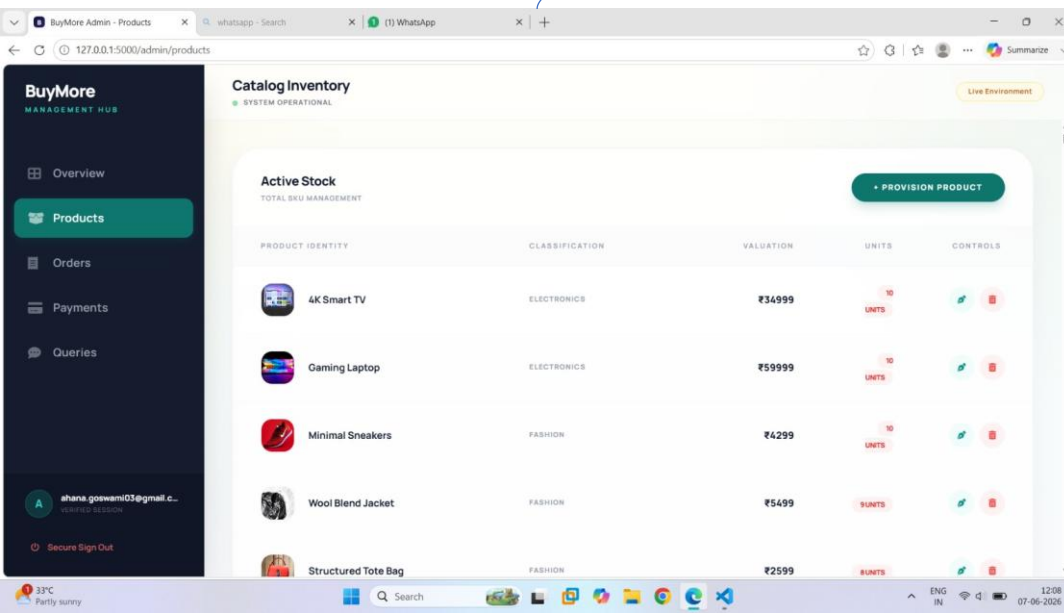
7.

Review Monitoring

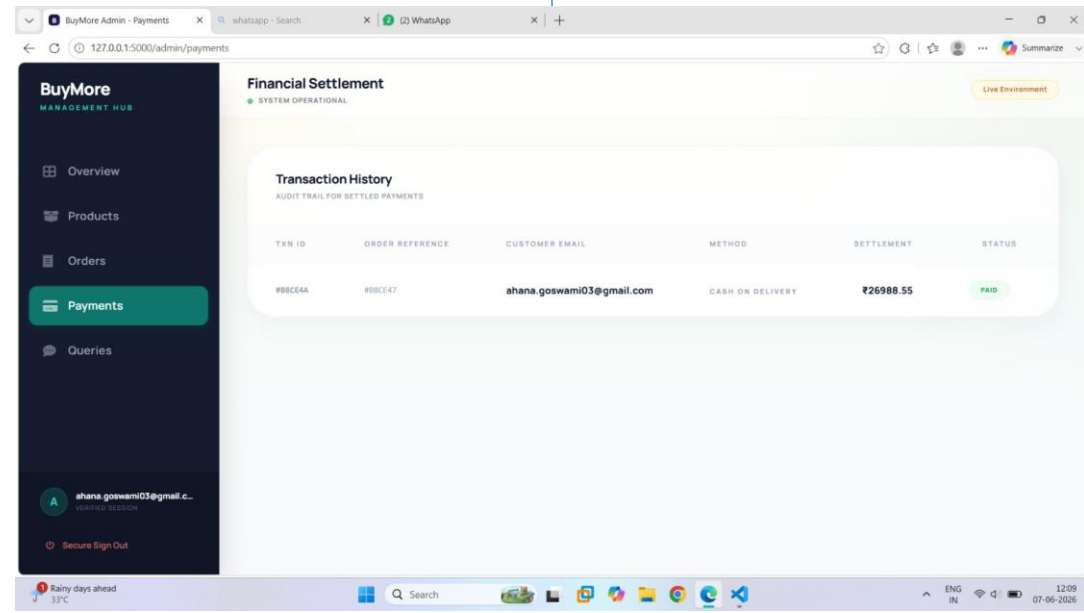


Admin Dashboard

Product Section

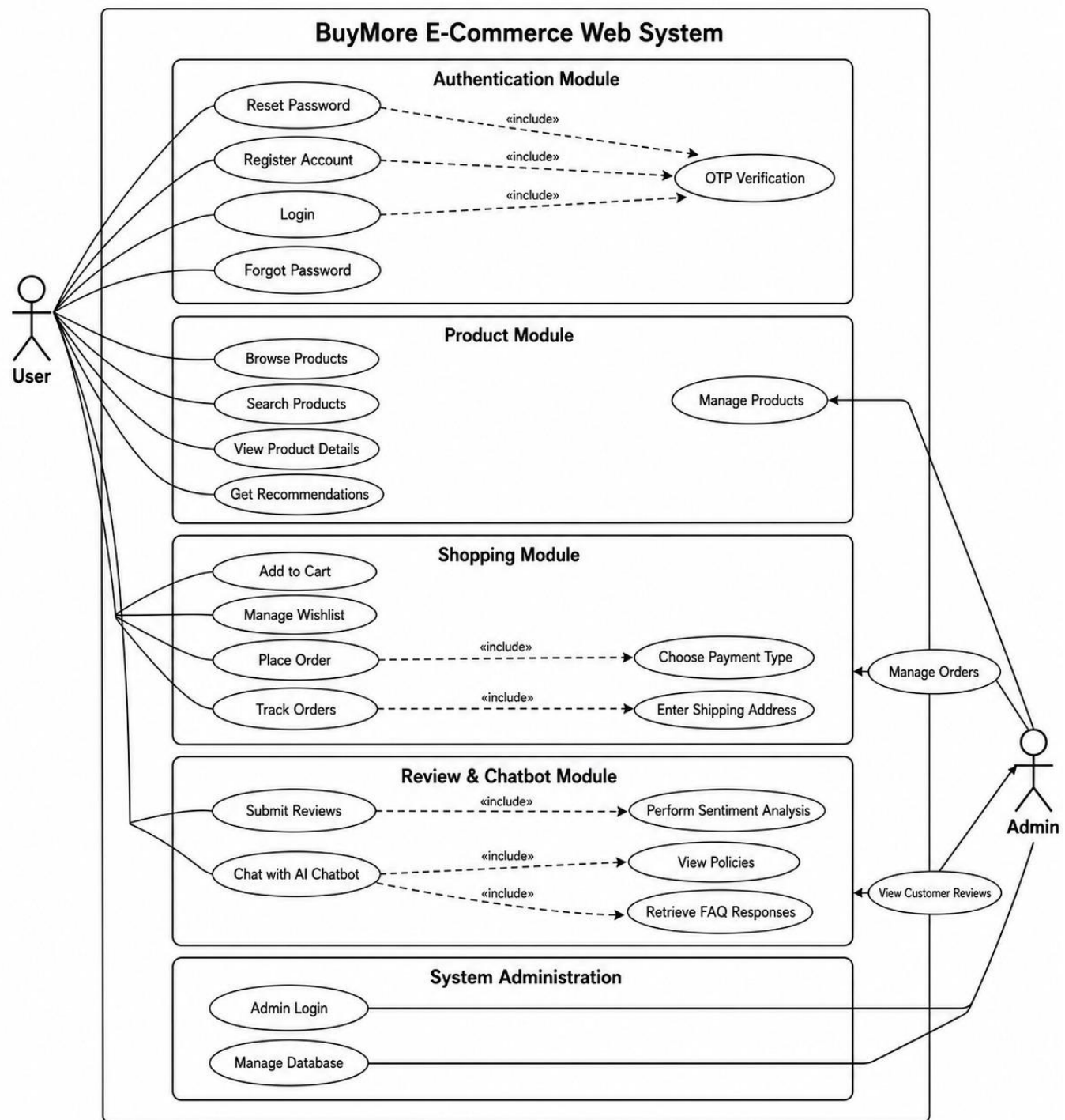


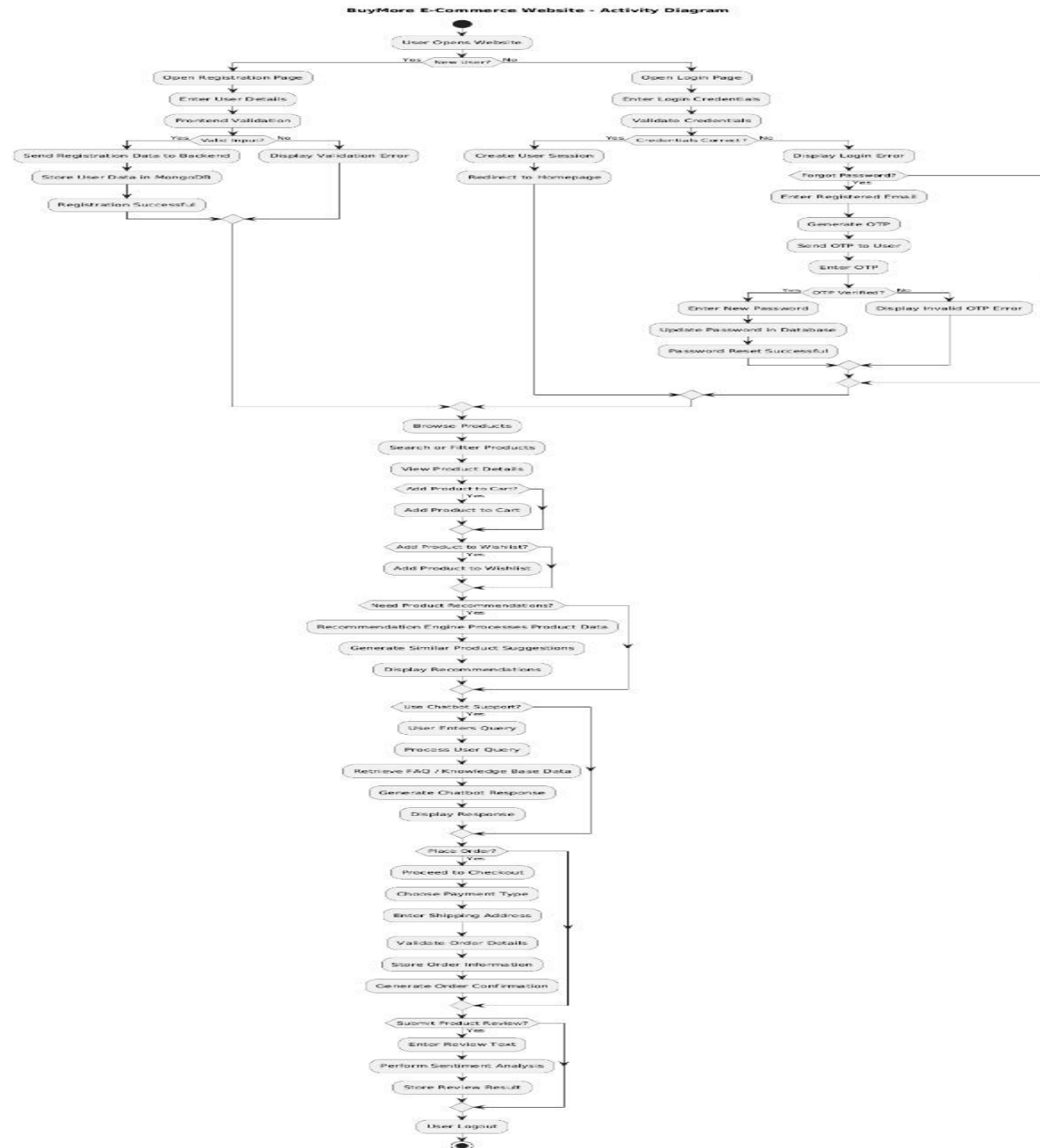
Payment Section



Admin Module Display

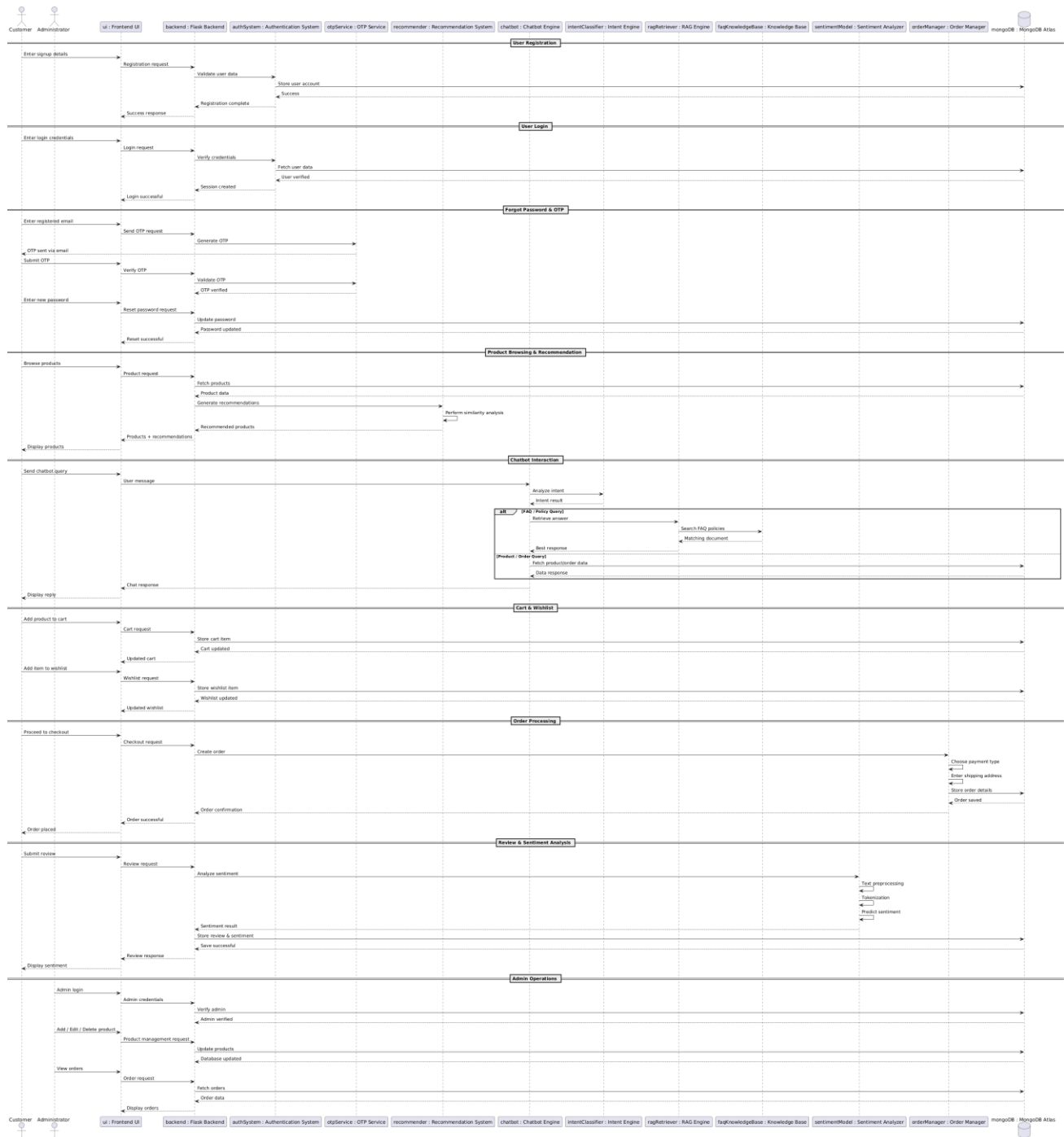
Use Case Diagram

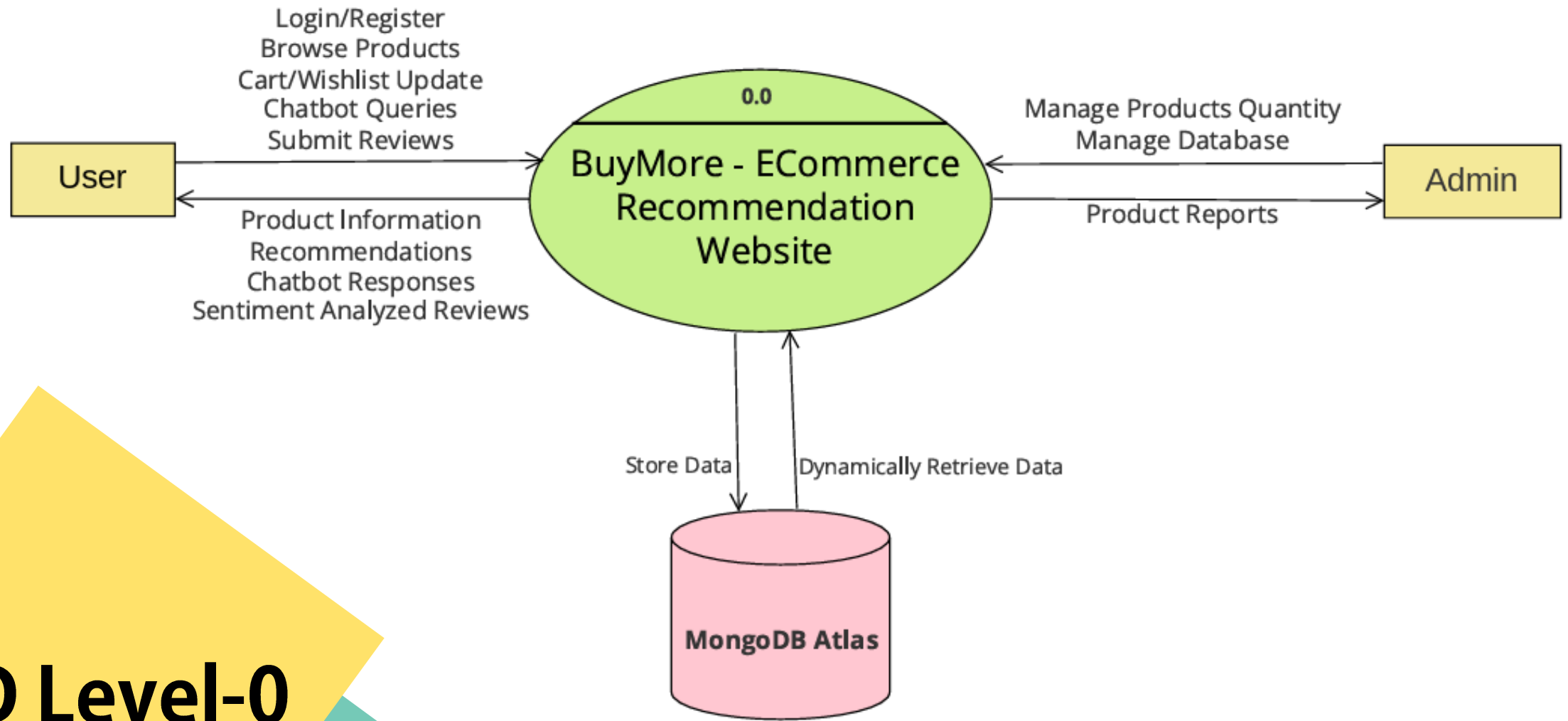




Activity
Diagram

Sequence Diagram

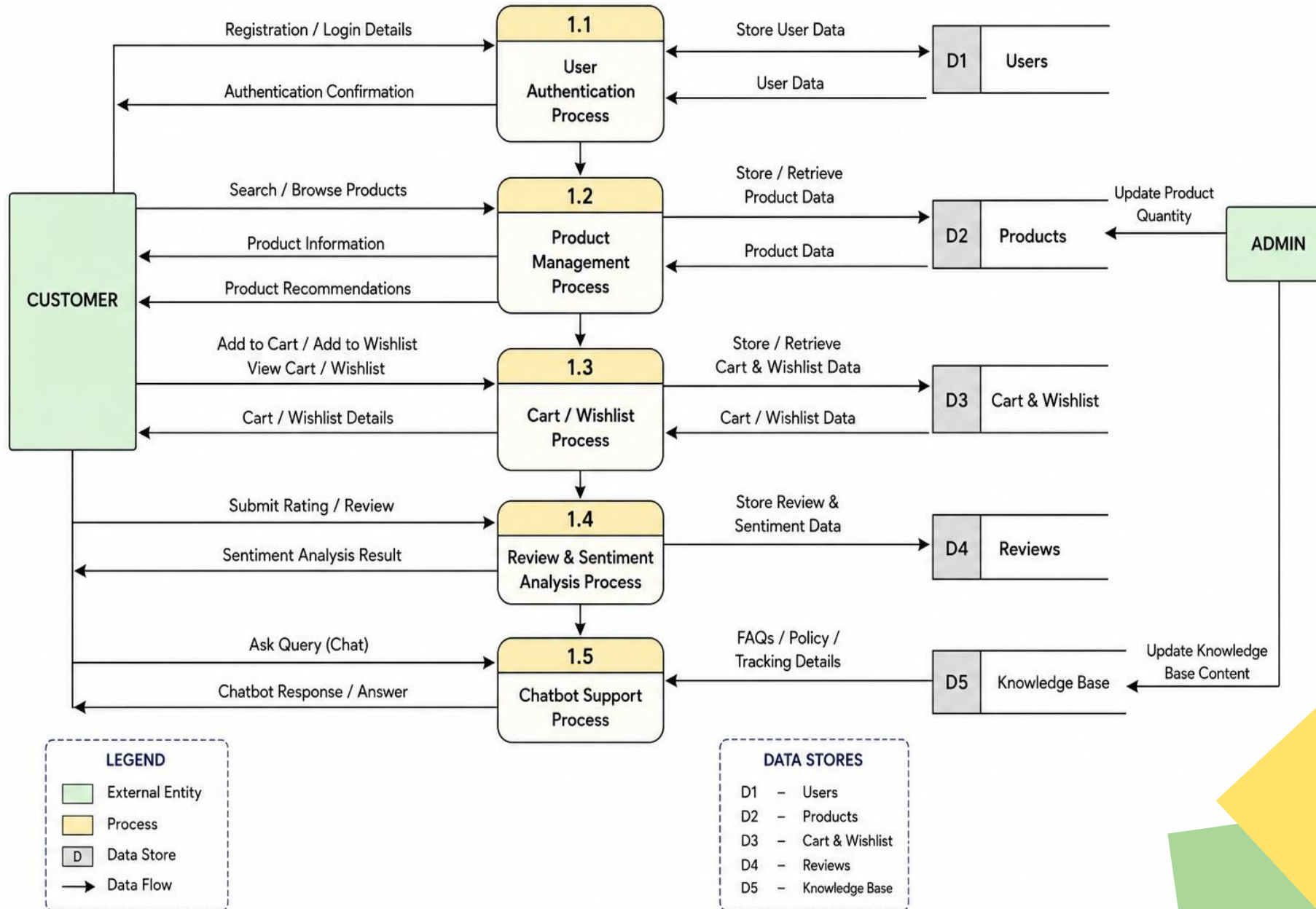




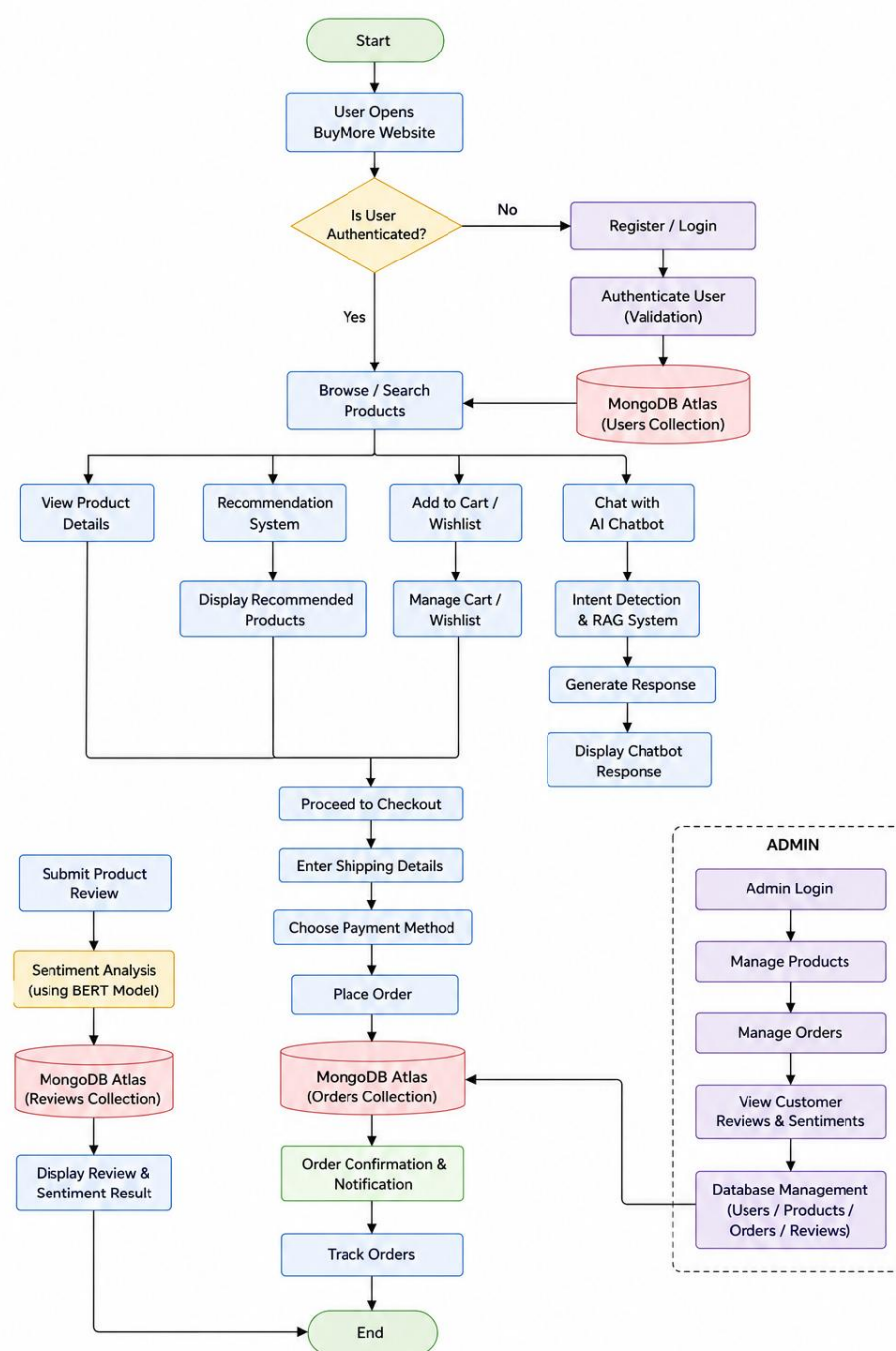
DFD Level-0 Diagram

DFD LEVEL 1 – BUYMORE E-COMMERCE WEB SYSTEM

(Gane and Sarson Notation)



DFD Level-1 Diagram



System Workflow

Customer Review & Sentiment Analysis

The system allows users to submit product reviews and ratings after purchasing products. Using NLP and Machine Learning techniques, the application analyses review text and classifies customer sentiment as Positive, Neutral, or Negative.

Features :

- Product Rating & Review System
- Real-Time Sentiment Classification
- Positive / Neutral / Negative Prediction
- Review-Based Recommendation Enhancement
- MongoDB Review Storage.



Smart Home Camera

electronics | INR 3,499

Designed for everyday reliability with business-grade quality, verified sellers, and secure delivery support.

User Reviews

★★★★★ 2.7/5 (3 reviews)

Raktima Biswas

★★★★★ 1.0

The Smart Home Camera did not fully meet my expectations. While the setup process was easy, the motion detection was sometimes inaccurate and notifications were delayed. The video quality was average, especially at night. Overall, the camera works for basic monitoring, but its performance could be more reliable.

NEUTRAL

Sourav Banerjee

★★★★★ 3.0

The Smart Home Camera performs its basic functions adequately. The video quality is acceptable for everyday monitoring, and the mobile app is fairly easy to use. Motion detection works most of the time, although there can be occasional delays in notifications. Setup is straightforward, but some advanced features require additional configuration. Overall, it is a decent option for basic home surveillance, though there is room for improvement in performance and reliability.

NEUTRAL

Ronnie Sarkar

★★★★★ 4.0

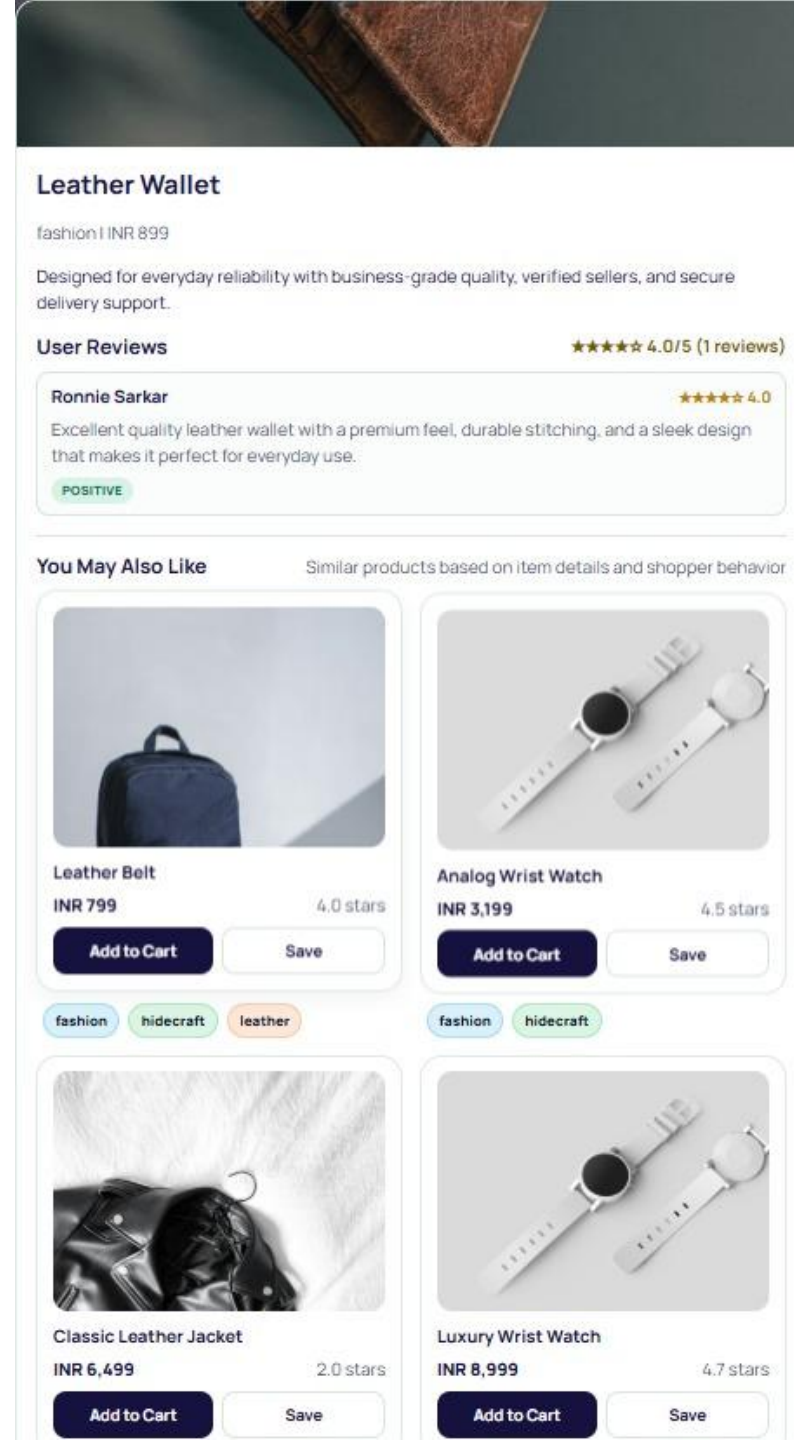
The Smart Home Camera has been an excellent addition to my home security setup. The video quality is clear, even in low-light conditions, and the motion detection feature works reliably. I especially appreciate the easy mobile app integration, which allows me to monitor my home from anywhere. Installation was quick and straightforward. Overall, it offers great value for money and provides peace of mind with its dependable performance and user-friendly features.

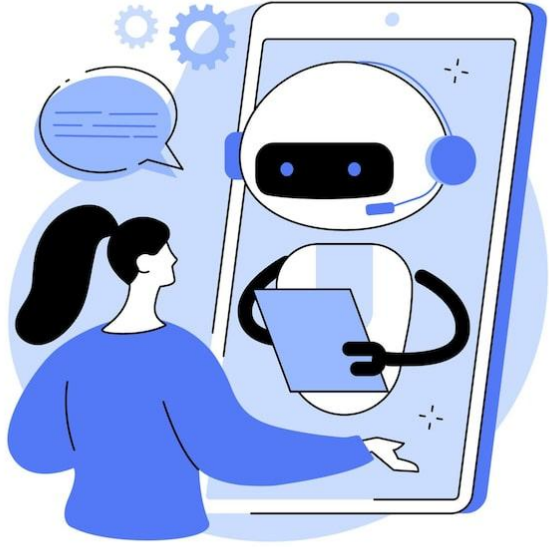
POSITIVE

Recommendation System

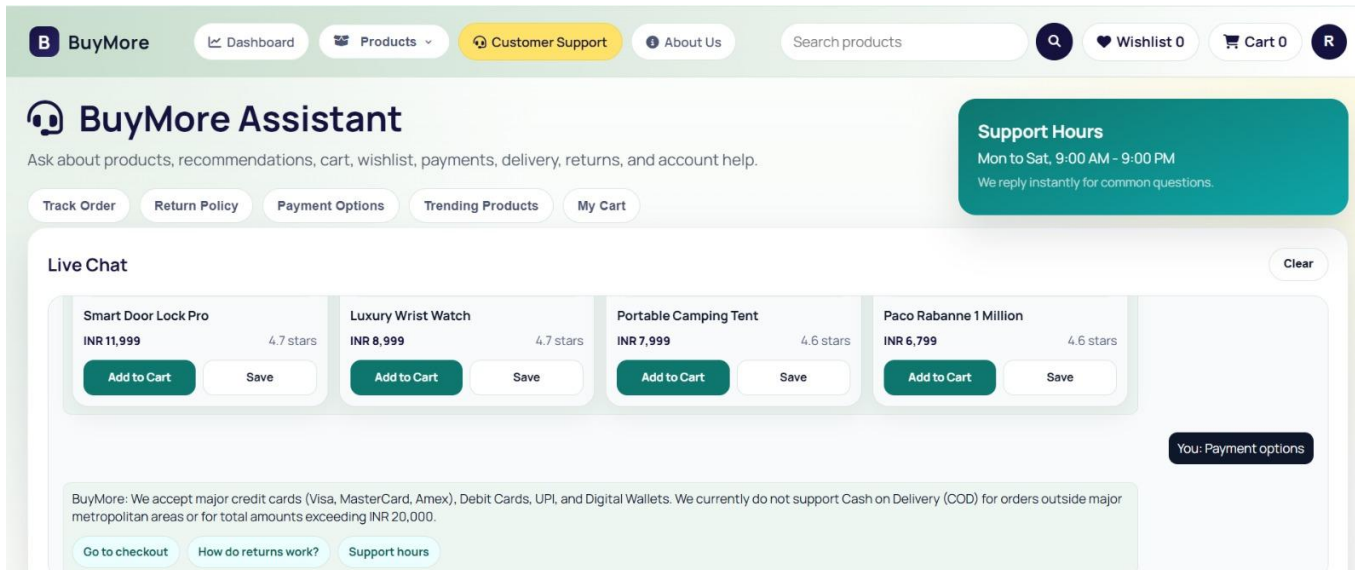


The **AI Recommendation Engine** is designed to deliver personalized product suggestions by leveraging Machine Learning techniques. It utilizes **TF-IDF Vectorization** to extract meaningful features from product information and **Cosine Similarity** to measure the relevance between products and user interests. The system analyzes multiple factors, including **product descriptions, product categories, customer reviews, and user preferences**, to understand product characteristics and user behavior. Based on this analysis, the recommendation engine generates **similar product recommendations** and **personalized suggestions**, helping users discover relevant products more efficiently and enhancing their overall shopping experience.





The **AI Chatbot Integration** module provides instant support and guidance to users throughout their shopping journey. It assists users with **product-related queries, cart management, FAQ handling, platform navigation, and order-related assistance**, ensuring a seamless and interactive user experience. The chatbot is powered by **Natural Language Processing (NLP)** techniques using **NLTK** and **Keras**, along with **Intent Classification** models that accurately understand user requests and deliver relevant responses. By providing real-time assistance and personalized interactions, the chatbot enhances user engagement, improves customer support efficiency, and simplifies the overall shopping process.



AI Chatbot

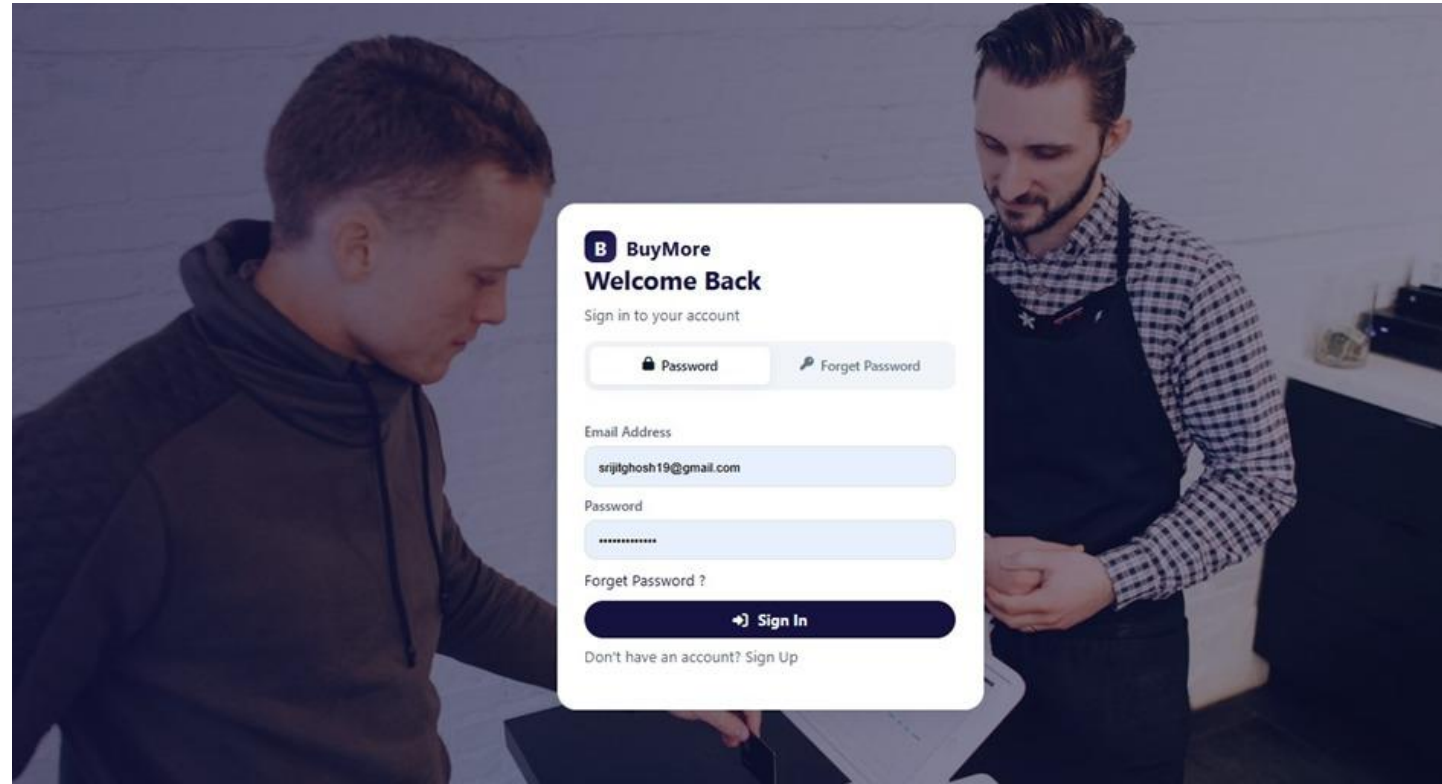
Authentication & Security

Security Features :

- OTP-Based Authentication
- bcrypt Password Hashing
- Secure Session Handling
- Role-Based Access Control
- Secure MongoDB Connection
- Environment Variable Protection

Validation Features :

- Email Validation
- Password Validation
- OTP Validation



MongoDB Atlas Integration :

BUYMORE uses MongoDB Atlas for secure cloud database management and real-time data handling.

Main Collections :

- users • products • cart • wishlist
- reviews • orders • admins

Database Features :

- CRUD Operations
- Real-Time Updates
- Secure Storage
- Session-Linked Cart
- Fast Data Retrieval



ecommerceDB +							
ecommercedb.jv3dfq4.mongodb.net > ecommerceDB							
>_ Open MongoDB shell + Create collection Refresh							
Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total Index size
admins	-	36.86 kB	698.00 B	5	139.00 B	1	36.86 kB
carts	-	36.86 kB	2.09 kB	4	523.00 B	1	36.86 kB
click_events	-	45.06 kB	42.56 kB	219	194.00 B	1	36.86 kB
contact_reports	-	36.86 kB	231.00 B	1	231.00 B	1	36.86 kB
orders	-	36.86 kB	839.00 B	1	839.00 B	1	36.86 kB
otp_codes	-	4.10 kB	0 B	0	0 B	1	4.10 kB
payments	-	36.86 kB	231.00 B	1	231.00 B	1	36.86 kB
products	-	151.55 kB	128.77 kB	305	422.00 B	1	36.86 kB
reviews	-	36.86 kB	6.13 kB	20	306.00 B	1	36.86 kB
users	-	36.86 kB	1.18 kB	7	168.00 B	2	73.73 kB
wishlists	-	36.86 kB	2.04 kB	4	509.00 B	1	36.86 kB

Mongo Database

Database Integration

Testing

Testing Performed :

Testing Type

Unit Testing

Integration Testing

System Testing

Security Testing

User Acceptance Testing

Purpose

Individual Functions

Module Communication

Complete Workflow

OTP & Sessions

User Experience



Testing Features :

- Login System
- Recommendation Engine
- Cart Operations
- Chatbot Responses
- Orders & Reviews



Deep Learning Recommendation Models

Mobile Application Development

Voice-Based Product Search

Payment Gateway Integration

Advanced AI Chatbot

Real-Time Analytics Dashboard

ERP Integration

Personalized Offers & Notifications

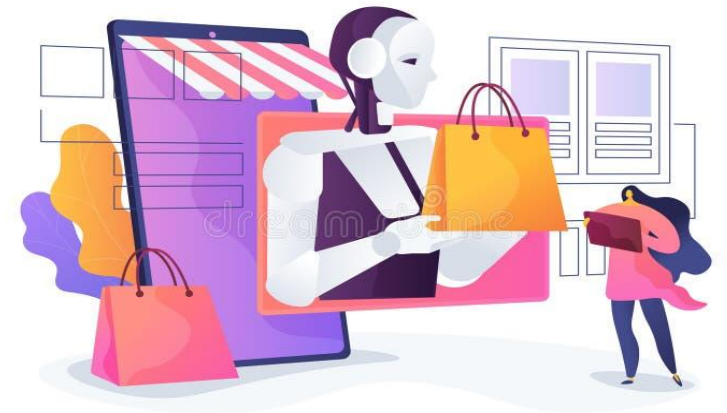


**Future
Scope**



- **BUYMORE** provides an intelligent ecommerce ecosystem.
- AI improves user engagement and shopping experience.
- Recommendation system enhances product discoverability.
- OTP authentication improves platform security.
- Chatbot increases customer support efficiency.
- Successfully integrates AI, Web Development, and Database Management

Conclusion



References



- [1] Flask Documentation. Available: <https://flask.palletsprojects.com/>
- [2] MongoDB Documentation. Available: <https://www.mongodb.com/docs/>
- [3] TensorFlow Documentation. Available: <https://www.tensorflow.org/docs>
- [4] NLTK Documentation. Available: <https://www.nltk.org/>
- [5] Scikit-learn Documentation. Available: <https://scikit-learn.org/stable/documentation.html>
- [6] Google Scholar, "Recommendation Systems Research Papers." Available:
<https://scholar.google.com/scholar?q=recommendation+systems+research+papers>

The background of the image consists of several concentric circles in shades of yellow, teal, and light green, creating a target-like pattern. The text "THANK YOU" is centered within the innermost yellow circle.

THANK YOU